

3.8 Maxima and Minima of Functions of Two Variables_Guide

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3.8 Maxima
and Mini...

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Definition:

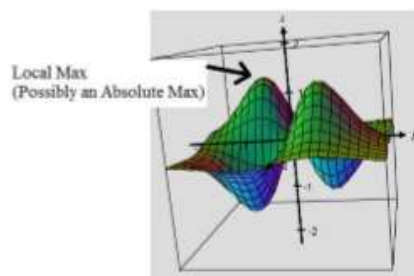
- A function f of two variables has a local maximum at (a, b) if $f(x, y) \leq f(a, b)$ when (x, y) is near (a, b) . The number $f(a, b)$ is called a local maximum value or simply a local maximum.
- A function f of two variables has a local minimum at (a, b) if $f(x, y) \geq f(a, b)$ when (x, y) is near (a, b) . The number $f(a, b)$ is called a local minimum value or simply a local minimum.

Note: We can make the notion of “near” precise by saying: “there exists some $\varepsilon > 0$ such that for every (x, y) in the disk centered at (a, b) of radius ε , $f(x, y) \leq f(a, b)$.”

Note: Local maxes/mins are z values!

Remark: If we replace the words “when (x, y) is near (a, b) ” in the above definitions with “for all (x, y) ” then we obtain the definition of absolute maximum and absolute minimum respectively, and we say f has an absolute maximum at (a, b) and f has an absolute minimum at (a, b) .

Ex: In the picture below a local max is identified. We cannot determine if it is a global max because it could be exceeded in a part of the graph that we cannot see.



Theorem: If a function of two variables f has a local maximum or minimum at (a, b) and the first-order partial derivatives of f exist there, then $f_x(a, b) = 0$ and $f_y(a, b) = 0$.

Proof: The proof is fairly straightforward; however, it is not included in the textbook.

Definition: A point (a, b) is called a critical point of $f(x, y)$ if $f_x(a, b) = 0$ and $f_y(a, b) = 0$ or if one of these partial derivatives does not exist.

Note: Critical points are where local minima/maxima MAY occur.

Example 1: Find all the critical points of the function $f(x, y) = x^2 + y^2 - 2x - 6y + 14$.

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We calculate $f_x = 2x - 2$ & $f_y = 2y - 6$

f_x & f_y are never undefined

$$\begin{cases} 2x - 2 = 0 \\ 2y - 6 = 0 \end{cases} \rightarrow \begin{matrix} x = 1 \\ y = 3 \end{matrix} \rightarrow (1, 3)$$

$(1, 3)$ is a critical point.

Theorem (Second Derivatives Test):

Suppose the second partial derivatives of f are continuous on a disk with center (a, b) , and suppose that $f_x(a, b) = 0$ and $f_y(a, b) = 0$. Let $D = D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$. Then,

1. If $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum. 
2. If $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum. 
3. If $D < 0$, then $f(a, b)$ is not a local maximum or local minimum. We say that (a, b) is a saddle point of f .
4. If $D = 0$, the test is inconclusive (i.e. it gives no information).

Proof: The proof is beyond the scope of this course. The key to proving this result is generalizing the notion of the second derivative test from calculus one.

Note: Results 1 and 2 are fairly easy to remember if one recalls the second derivative test from calculus 1.

Example 2: Find the local minimum and maximum value(s) and any saddle points of the function

$$f(x, y) = x^4 + y^4 - 4xy + 1.$$

We calculate $f_x = 4x^3 - 4y$ & $f_y = 4y^3 - 4x$. These are never undefined.

$$\begin{cases} 4x^3 - 4y = 0 & \textcircled{1} \\ 4y^3 - 4x = 0 & \textcircled{2} \end{cases}$$

By $\textcircled{1}$ $4y = 4x^3 \rightarrow y = x^3$

By $\textcircled{2}$ $4(x^3)^3 - 4x = 0 \rightarrow 4x^9 - 4x = 0 \rightarrow 4x(x^8 - 1) = 0$

$$4x = 0 \text{ or } x^8 - 1 = 0$$

$$x = 0 \text{ or } x = \pm 1$$

So $(0, 0)$, $(-1, -1)$ & $(1, 1)$ are critical pts.

$$f_{xx} = 12x^2$$

$$f_{yy} = 12y^2$$

$$f_{xy} = -4$$

$$D(0,0) = f_{xx}(0,0) \cdot f_{yy}(0,0) - [f_{xy}(0,0)]^2$$

$$= (0) \cdot (0) - [-4]^2 = -16 \quad \text{Saddle pt at } (0,0)$$

$$D(-1,-1) = (\underline{12}) \cdot (12) - [-4]^2 > 0 \quad f_{xx} > 0 \quad \cup \quad \text{Local min of } -1 \text{ @ } (-1,-1)$$

$$D(1,1) = (\underline{12})(12) - [-4]^2 > 0 \quad f_{xx} > 0 \quad \cup \quad \text{Local min of } -1 \text{ @ } (1,1)$$

Note: Language in the above example is very important. In particular, it is proper to say that there is a saddle point at $(0,0)$. It is also proper to say that there is a local minimum of -1 at $(1,1)$, or there is a local minimum of -1 at $(-1,-1)$. It would NOT be correct to say that there is a local minimum of $(1,1)$.

Example 3: Consider the function $f(x,y) = x^3 - y^3$. Show that $(0,0)$ is a critical point of f . Then, show that the second derivatives test is inconclusive in classifying $(0,0)$.

Note that $f_x = 3x^2$ & $f_y = -3y^2$. These are never undefined.

$$\begin{cases} 3x^2 = 0 \\ -3y^2 = 0 \end{cases} \rightarrow \begin{matrix} x=0 \\ y=0 \end{matrix} \rightarrow (0,0) \text{ is a critical point.}$$

$$f_{xx} = 6x \quad f_{yy} = -6y \quad \& \quad f_{xy} = 0$$

$$D(0,0) = f_{xx}(0,0) \cdot f_{yy}(0,0) - [f_{xy}(0,0)]^2 = (0)(0) - [0]^2 = 0$$

The test tells us nothing.

We now move to discussing absolute extrema.

Definition (intuitive): A closed set in $\mathbb{R}^2$ is one that contains all of its boundary points.

Note: For now, we will think of "boundary" in the intuitive sense. It is possible however, to give a mathematically rigorous definition of boundary point.

Ex: Based upon the above definition $\{(x, y) : x^2 + y^2 \leq 1\}$ is a closed set, and $\{(x, y) : x^2 + y^2 < 1\}$ is not a closed set.



Definition: A bounded set in $\mathbb{R}^2$ is one that is contained within some disk.

Theorem (Extreme Value Theorem for Functions of Two Variables):

If f is continuous on a closed, bounded set D in $\mathbb{R}^2$, then f attains an absolute maximum value $f(x_1, y_1)$ and an absolute minimum value $f(x_2, y_2)$ at some points (x_1, y_1) and (x_2, y_2) in D .

Proof: Beyond the scope of this course. This theorem is often proven in Real Analysis or General Topology.

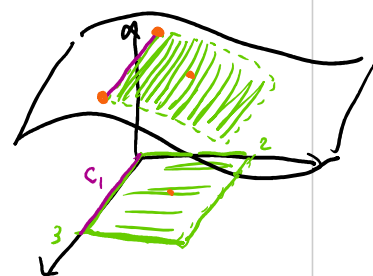
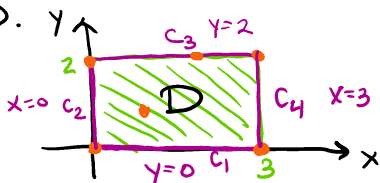
Steps for Finding the Absolute Max or Min on a Closed Bounded Set:

1. Confirm that f is continuous.
2. Find the values of f at the critical points of f in D .
3. Find the extreme values of f on the boundary of D .
4. The largest of these values is the absolute maximum of f on D . The smallest of these values is the absolute minimum of f on D .

Example 4: Find the absolute maximum and minimum values of the function $f(x, y) = x^2 - 2xy + 2y$ on the rectangle $D = \{(x, y) : 0 \leq x \leq 3 \text{ and } 0 \leq y \leq 2\}$. Also, state where the absolute maximum and minimum values occur.

Step 1: $f(x, y)$ is a cont. polynomial function.

D is closed & bound, so by the EVT $f(x, y)$ has an abs. min & abs. max on D .



$$f_x = 2x - 2y \text{ \& } f_y = -2x + 2$$

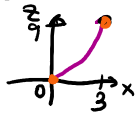
Step 2: ...

$$f_x = 2x - 2y \text{ \& } f_y = -2x + 2$$

Step 2: $\begin{cases} 2x - 2y = 0 \text{ (1)} \\ -2x + 2 = 0 \text{ (2)} \end{cases} \rightarrow \text{by (2) } x=1 \text{ \& then by (1) } y=1 \rightarrow (1,1) \text{ is a critical pt.}$
 $\& f(1,1) = 1$

Step 3: We now check along the boundary of D for other extreme points.

Along C_1 our function becomes $f(x,y) = f(x,0) = x^2$ for $0 \leq x \leq 3$.



$$f_x(x,0) = 2x \quad f_x(x,0) \text{ is never und. \& } f_x(x,0) = 0 \text{ when } x=0$$

We also need to find what happens at $x=0$ \& $x=3$

$$f(0,0) = 0 \quad \& \quad f(3,0) = 9$$

Along C_2 our function becomes $f(x,y) = f(0,y) = 2y$ for $0 \leq y \leq 2$.



$$f_y(0,y) = 2 \quad f_y(0,y) \text{ is never und. \& never equals 0.}$$

We also need to find the behavior at $y=0$ \& $y=2$.

$$f(0,0) = 0 \quad \& \quad f(0,2) = 4$$

Along C_3 our function becomes $f(x,y) = f(x,2) = x^2 - 4x + 4$ for $0 \leq x \leq 3$.

$$f_x(x,2) = 2x - 4, \quad f_x(x,2) \text{ is never und. \& equal 0 when } x=2.$$

We also need to find the behavior at $x=0$ \& $x=3$

$$f(2,2) = 0 \quad f(0,2) = 4 \quad f(3,2) = 1$$

Along C_4 our function becomes $f(x,y) = f(3,y) = 9 - 4y$ for $0 \leq y \leq 2$

$$f_y(3,y) = -4 \quad \text{This is never und. or equal to 0.}$$

We just check at the endpoints of $y=0$ \& $y=2$

$$f(3,0) = 9 \quad f(3,2) = 1$$

The absolute max of $f(x,y)$ over D is 9 (occurs at $(3,0)$)

The absolute min of $f(x,y)$ over D is 0 (occurs at $(0,0)$ \& $(2,2)$)